

Paired and Two-Sample t -Test Examples

CSU, Chico Math 351

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Identifying Paired vs. Two-Sample t-Tests (1)

The following examples are very similar in content and structure, but one uses a paired t-test and the other uses a two-sample t-test. Always look carefully at the data to see if the following criteria are met:

1. Are the two variables intimately connected (and NOT independent)?
2. Does each single entry from the first group correspond to exactly one entry in the other group?
3. Are you testing the average difference between EACH corresponding pair in the two groups?

If all of the above apply, use a paired t-test.

Identifying Paired vs. Two-Sample t-Tests (2)

On the other hand:

1. Are the two variables independent?
2. Are you testing the difference of means between the two groups?

If these are true instead, use a two-sample t-test.

Comparing Class Scores, Case 1

A university is comparing the performance of students taking an undergraduate course. Two classes of 30 students each had their final course grades recorded after one semester. Their scores are recorded in the "classes.csv" file, available in the Week 9 Folder. Obtain a confidence interval and perform a hypothesis test to determine whether there was a significant difference in course grades among the two classes.

Comparing Class Scores, Case 2

A university is testing out a new study module for an undergraduate course. As a trial run, 20 students took an exam before taking the module, and took a similar exam after taking the module. Their scores are recorded in the "scores.csv" file, available in the Week 9 Folder. Obtain a confidence interval and perform a hypothesis test to determine whether there was a significant change in exam scores after taking the module.

Which Test?

Which of these cases requires a paired t-test? Which requires a two-sample t-test?

Conclusions

What are your conclusions for each case? How would you explain them to a non-statistician (without using reject/fail to reject H_0)?

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Examples

References

references I

David M Diez, Christopher D Barr, and Mine Cetinkaya-Rundel. *OpenIntro Statistics*. CreateSpace independent publishing platform, third edition, 2015.

Hadley Wickham. *ggplot2: elegant graphics for data analysis*. Springer Science and Business Media, 2009.

Adapted from notes by Dr. Roualdes.